Franoll Fantu

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Education

Simon Fraser University Graduation: Apr 2028
Bachelor of Sciences in Computer Science *Burnaby, BC*

Work Experience

Software Developer Co-op May 2026 – Present
Shared Services Canada - CanAI Garage *Vancouver, BC*

- Collaborated with a cross-Canada team to build a **predictive AI fleet management system**
- Developed and tuned **regression and classification models** for fleet forecasting and integrated them into production workflows
- Implemented **MLflow** to track model parameters, metrics, and experiments across **ML pipelines**
- Built **RESTful APIs** with **Django/Python** and a **Redis** caching system, reducing average API response time by **50%**
- Containerized application services using **Docker** and deployed them to **Azure VMs** through **CI/CD pipelines**
- Served as Scrum Master, managing sprints in **Jira**, coordinating development across shared VM environments, and communicating progress to government stakeholders

Machine Learning Intern January 2026 – April 2026
VJDS International Inc. *Calgary, Alberta*

- Built **ML pipelines** for classification, enrichment, and predictive scoring integrated into production services
- Designed end-to-end **data pipelines** for ingestion, preprocessing, validation, and feature transformation using **Python**
- Developed backend services and **RESTful APIs** using **FastAPI** and **Python**
- Implemented **Redis** caching and storage for API requests, reducing response times by **75%**
- Performed **spatial analysis** to compute density metrics and generate heatmap visualizations for trend analysis

Projects

PokéVault | *Java, Spring Boot, PostgreSQL, Docker, Spring Data JPA, Flyway, Testcontainers* **[GitHub] [Demo]**

- Built a **Spring Boot** REST API for storing and retrieving Pokémon TCG collection data using **Java 21**
- Integrated **PostgreSQL**, **Spring Data JPA**, and **Flyway** for persistent storage and version-controlled database migrations
- Containerized the database with **Docker Compose** and implemented automated testing with **JUnit**, **Mockito**, and **Testcontainers**

AI Polypharmacy Agent | *TechNation × TELUS Hackathon* | *React, Python, FastAPI, PostgreSQL, OCR* **[GitHub] [Demo]**

- Built a **Flask** RAG pipeline using **ChromaDB** vector search over 10,000+ FDA drug interaction records, grounding **Gemini** responses in verified clinical data
- Integrated **PaddleOCR** and **spaCy NER** to extract medications from prescription images while filtering sensitive information
- Automated citation-backed PDF report generation in 5 seconds using vector similarity, enabling real-time decision support

QuizLing | *React, Node.js, JavaScript, GeminiAI, Firebase, Figma* **[GitHub] [Demo]**

- Led a **4-person Agile Scrum team**, managing bi-weekly sprints in **Jira** and improving delivery consistency by **25%**
- Built a responsive **React + Tailwind** frontend and integrated **REST APIs** for translation and text-to-speech
- Designed and maintained a **Firebase Firestore** schema for user progress, quizzes, achievements, and real-time state

Leadership

Vice President, Web & Digital Systems January 2026 – Present
Engineering Without Borders Canada (SFU Chapter) *Burnaby, BC*

- Architected, maintained, and optimized the organization's public-facing website
- Developed and deployed frontend features and pages using **HTML**, **CSS**, **JavaScript**, and **React**
- Managed site structure, content updates, and deployment workflows to support events, recruitment, and project launches
- Collaborated with executive leadership to align technical decisions with long-term digital strategy

Software Engineer January 2026 – June 2026
Enactus SFU *Burnaby, BC*

- Led development of a healthcare application using **React Native** with modular, component-based architecture
- Integrated cloud authentication, data storage, and deployment workflows using **AWS Amplify** and **AWS Cognito**
- Worked closely with design and strategy leads to translate product requirements into scalable technical solutions
- Defined frontend architecture, conducted code reviews, and enforced engineering best practices across the team
- Planned development sprints and decomposed features into clear, well-scoped engineering tasks

Technical Skills

Programming & Query Languages: Java, Python, C, C++, JavaScfipt, TypeScript, Bash, PHP
Data & Analytics Tools: Excel, Power BI, Tableau, pandas, NumPy, Matplotlib
Frameworks & Platforms: React, Node.js, Express.js, Flask, FastAPI, Firebase
DevOps & Collaboration: Linux, UNIX, Docker, Git, GitLab, Jira, Confluence
Concepts: Agile Scrum, QA Testing, ETL Pipelines, Data Validation, Dashboard Development, Technical Documentation